

Code-Layout, Readability and Reusability

Date	15 March 2026
Team ID	XXXXXX
Project Name	Rising Water
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	4 spaces used uniformly across all files for better readability
2	Proper File Structure	Files and folders are logically organized	No	Files are currently in single folder. Need to separate into src, utils, assets folders
3	Meaningful Variable Names	Variables reflect their purpose clearly	Partial	Most variables like userEmail, password are clear. Some temp variables like a, b need renaming
4	Function / Method Names	Functions are descriptively named	No	Some functions use generic names like fun1(), doStuff(). Need descriptive names
5	Code Comments	Inline and block comments explain logic	Partial	Key functions have comments. Complex logic sections need more inline comments
6	Modular Design	Code is split into reusable functions/modules	No	Code is in single file. Plan to split into reusable modules for Registration, Login, Validation
7	No Redundant Code	Duplicate or unused code is removed	Yes	Duplicate code removed. Common functions reused across modules
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Try-catch blocks used. Invalid inputs show user-friendly error messages

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology Where Reused		Reusability Level (High / Medium / Low)
1	Email Validation Module	HTML, JavaScript, React	Validates email format during Registration & Login	High

2	Password Encryption Utility	Java / Python	Encrypts and decrypts user passwords using bcrypt	High
3	Social Login API Handler	OAuth 2.0, Facebook API, Gmail API	Handles login/register via Facebook and Gmail	Medium
4	Confirmation Email Template	HTML, CSS	Reused for registration confirmation and password reset	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	4	Code is well organized into separate modules for Registration, Login, and Utilities. Proper indentation and naming conventions followed.
Readability	5	Variable names are meaningful. Code is easy to understand with proper spacing and consistent formatting.
Reusability	4	Common functions like email validation and password encryption are reused across multiple modules.
Documentation / Comments	3	Important functions have inline comments. README file explains setup and usage. Can add more comments for complex logic.
Overall Score	4	Overall code quality is good with clean structure and reusability. Minor improvements needed in documentation.